



ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

PROJECT REPORT

Smart Classroom

An Automated Occupancy-Sensing and UPS-Backed Power Management System

Report Details

Group	STAT IC
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Course No & Name	EEE-4204, Electronics-I Lab(HW)
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Prepared By

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1. Abstract

This project presents the design, Proteus simulation, and physical breadboard prototyping of an intelligent, energy-efficient 'Smart Classroom' system equipped with automated seat-occupancy sensing and uninterrupted power management. The system integrates a 6-seat occupancy detection network utilizing mechanical microswitches, a 1-BJT touch-override system, and an automatic Uninterruptible Power Supply (UPS) switching circuit. Occupancy sensing is handled by six microswitch-driven BC547 transistor stages that OR-wire onto a shared occupancy bus through steering diodes, with an NE555 monostable timer providing a short delay-off grace period, driving BJT-and-relay output stages that manage 12V classroom loads (miniature brushless cooling fans and LED light modules). During main power loss, a 12V 5-pin SPDT relay seamlessly drops heavy loads while switching the 5V operational logic to a 7.4V backup battery via an LM7805 voltage regulator, maintaining total system logic integrity. Due to time constraints, the IR door entry/exit sensor and the NTC thermistor-based fire alarm modules were left incomplete as Proteus schematics and were not finished or carried over into the physical breadboard build. A separate, self-contained touch-override sub-circuit was also modeled on the same schematic sheet; it is functionally independent of the door sensor and fire alarm blocks. Despite hardware component failures during physical assembly (specifically faulty BJT batches), hardware troubleshooting methodologies enabled a successful migration to an optimized 1-BJT override circuit. The complete system was assembled and tested across two interconnected breadboards, proving highly practical for modern automated building management.

2. Introduction and Objectives

2.1 Background and Motivation

Educational institutions frequently suffer from significant electrical energy waste caused by unmanaged classroom loads — such as lighting and ceiling ventilation fans left running in empty rooms or unoccupied seating rows.

To resolve this challenge, this project implements a Smart Classroom prototype that automates load shedding based on seat occupancy, combined with an uninterrupted backup power topology so that control logic survives unexpected power blackouts.

2.2 Design Objectives

- Design and construct a multi-seat occupancy sensing network using microswitches driving individual BC547 transistor stages, OR-wired onto a shared occupancy bus with an NE555 monostable delay-off timer, to activate 12V ventilation fans and LED lighting modules.
- Develop a reliable manual touch-override system (utilizing an optimized 1-BJT configuration) to allow instructor/manual control over room electronics.

- Engineer a seamless relay-based UPS power management architecture that disconnects heavy 12V loads during power outages while keeping 5V logic powered through a 7.4V backup battery and LM7805 regulator.
- Implement dual KCD11 rocker switches: SW1 to simulate a main power blackout, and SW2 in series with the backup battery to allow it to be manually isolated from the circuit.

3. System Architecture and Simulation

3.1 Integrated Schematic Overview

The overall circuit design was initially modeled and validated using Proteus 8 Professional software. Signal flow travels sequentially from input transducers (Microswitches, Touch Plate) to signal conditioning blocks (BC547 occupancy transistor stages, the NE555 delay-off timer, and the touch-override BJT driver), driving the 12V output relays.

FIGURE: FULL SYSTEM SCHEMATIC (PROTEUS 8)

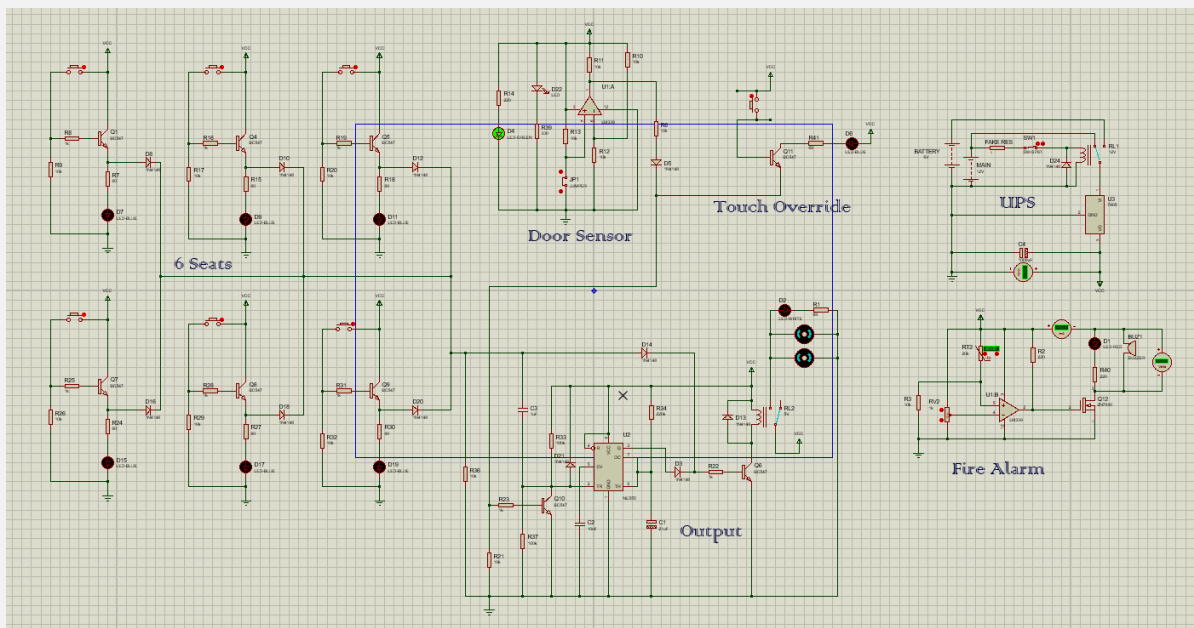
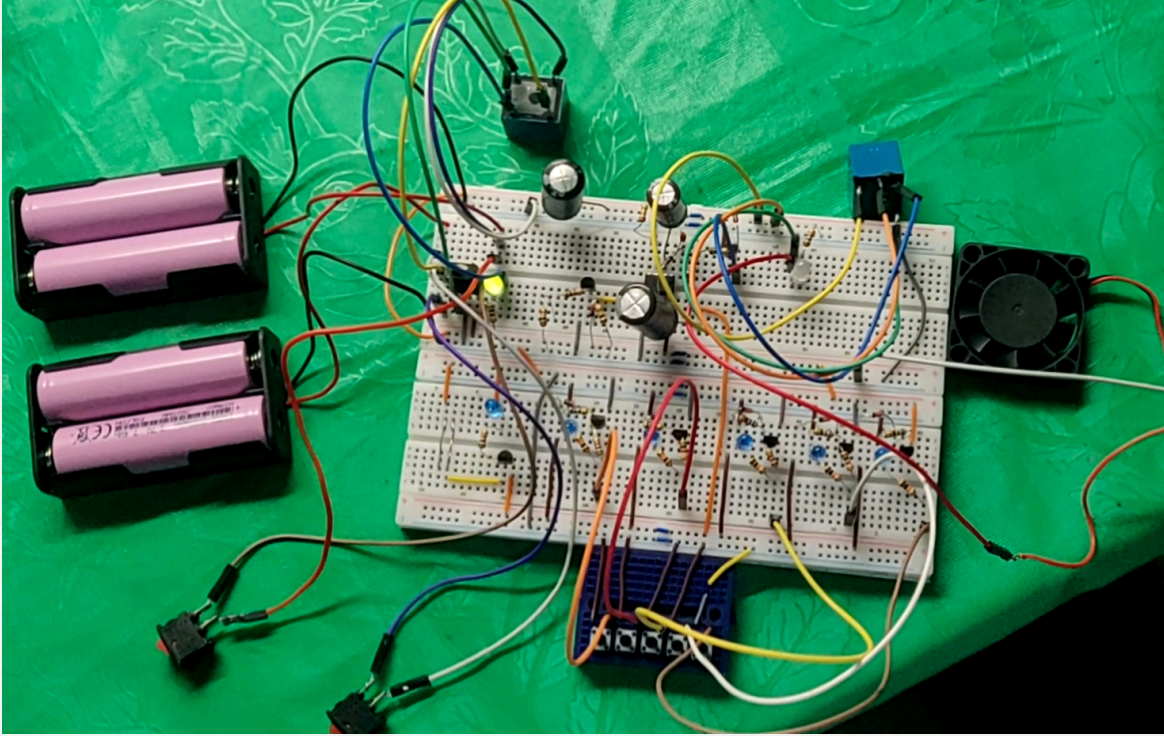


FIGURE: FULL SYSTEM SCHEMATIC (HARDWARE PHOTO)



3.2 Simulation Analysis and Transient Behavior

During Proteus transient simulation, flipping the main power switch (SW1) to simulate a grid blackout triggers a SPICE solver warning: '[SPICE] transient GMIN stepping at time=t'. This warning represents a standard mathematical convergence algorithm executed by SPICE when handling instantaneous voltage transitions across inductive relay coils (RL1) and switching diodes. It confirms that the simulation software successfully adjusted minimum conductance steps to resolve non-linear differential equations without crashing. In the physical breadboard implementation, physical contact arcing and diode snubbing (1N4007 flyback diodes) absorb these transient voltage spikes naturally.

4. Module-Level Design and Operating Principles

4.1 Occupancy Sensing and Delay-Off Timer Module (Microswitches, BC547 Transistors, NE555 Timer)

Each of the 6 seats is fitted with a lever microswitch feeding a dedicated BC547 transistor stage (e.g. Q8, Q9), each with its own base/collector resistor network (R27–R32). When a seat is occupied, its switch closes and turns that seat's transistor on, which lights a blue indicator LED (D18, D19, ...) showing the

seat is occupied and pulls a shared “occupancy bus” line active. All six seat stages are OR-wired onto this common bus through steering diodes (D20, D14, etc.), so the bus goes active as soon as any single seat is occupied and only goes inactive once every seat is empty. The occupancy bus directly drives the output transistor (Q6) through R22, which switches relay RL2 to energize the 12V loads (the cooling fan and LED module). As long as at least one seat remains occupied, the bus stays high and the output relay stays continuously latched ON, with no timing involved at this stage.

When the last occupied seat empties, the bus transitions from active to inactive; this falling edge is coupled through the R36/R37/C2/Q10 sensing network into the trigger pin (pin 2) of an NE555 timer (U2) wired as a monostable multivibrator. The trigger fires a single output pulse at pin 3, sized by the R34/C1 timing network to last approximately 5 seconds, during which the 555's output keeps Q6 (and therefore RL2 and the load) energized even though the seat bus itself has gone low. Once the monostable period elapses, pin 3 drops low, Q6 turns off, RL2 de-energizes, and the load switches off — giving the classroom a 5-second grace period after the last occupant stands up before the fan/lights actually cut out. Net logic: Output = (any seat occupied) OR (5-second monostable timer running, triggered on the occupied-to-vacant transition).

FIGURE: OCCUPANCY SENSING MODULE (PROTEUS SNIP)

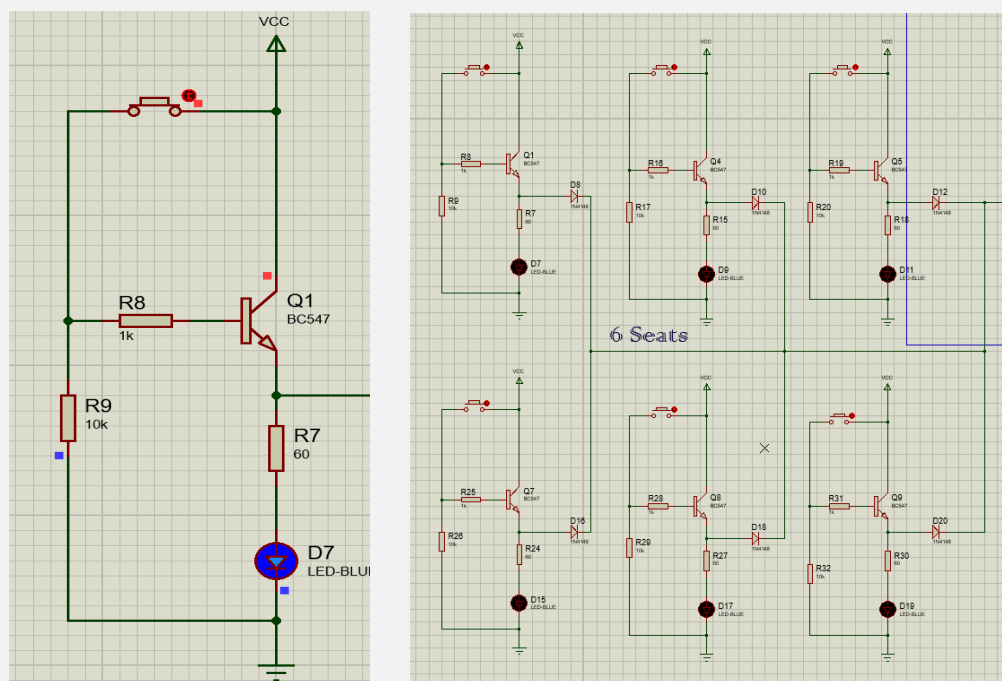


FIGURE: OCCUPANCY SENSING MODULE (HARDWARE PHOTO)

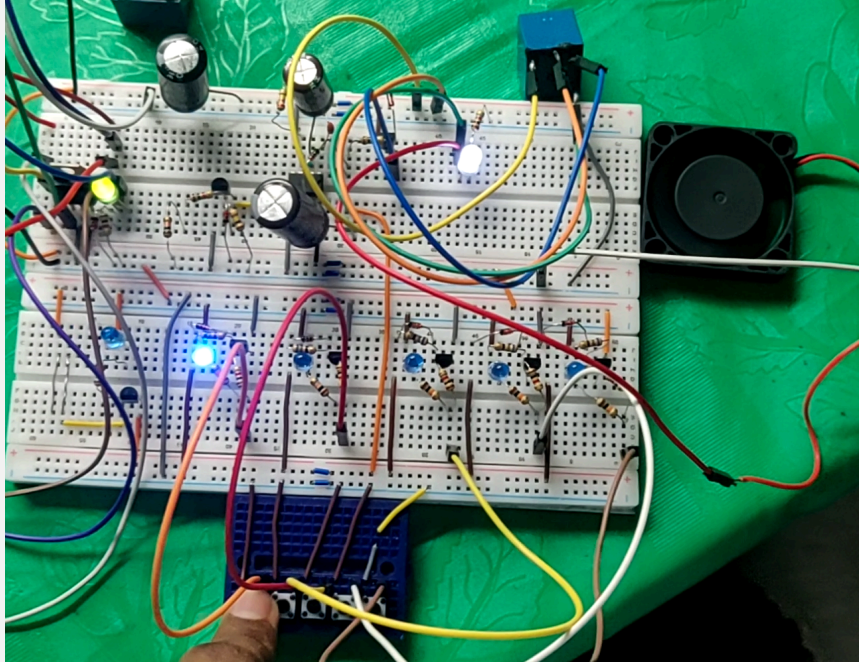
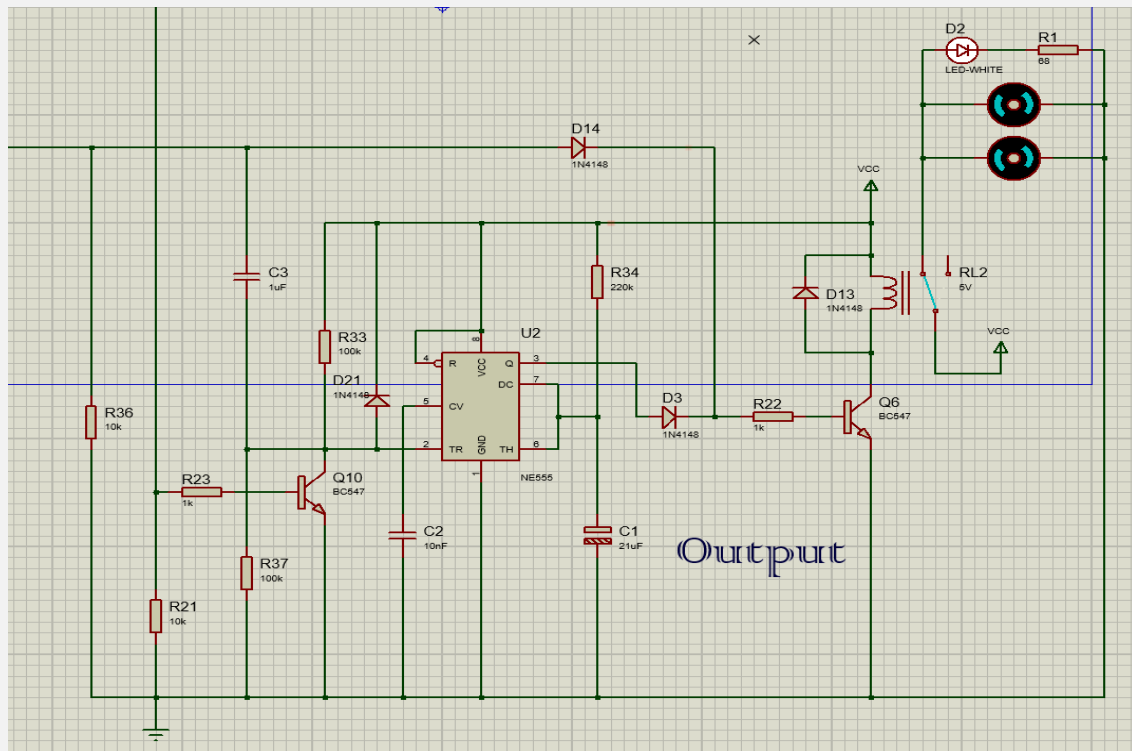


FIGURE: OUTPUT MODULE(PROTEUS SNIP)



4.2 Touch-Override Circuit (Single-BJT Design)

To allow manual control over the automated load switching, a touch-override circuit is incorporated. While an initial multi-transistor flip-flop design suffered from component defects, the final optimized design uses a single BC547 BJT. The transistor's collector is connected to the output driver's source line, and its emitter is tied to the positive bus. The base line is biased toward the source through a resistor, with a deliberate break left in this line — the two ends of the break are left exposed as bare touch points. In the resting state, with the exposed wires disconnected, no base current flows and the transistor remains OFF. When a user touches both exposed wire ends simultaneously, the finite resistance of human skin completes a weak current path from the bus into the base terminal. This small bias current is enough to switch the transistor fully ON, overriding the sensor inputs and forcing the output relay into a continuous active state.

The schematic labels the transistor Q11 (BC547), with its emitter tied to the positive supply rail and its collector driving up through R41 (68Ω) into indicator D6 (blue LED) back to VCC, mirroring the collector/emitter arrangement described above — D6 and R41 stand in for the real output-driver/load line the override forces active. The exposed touch points are modeled by a 10kΩ resistor labeled "HUMAN SKIN," representing the same bare-wire finger-touch resistance used in the physical build.

FIGURE: TOUCH-OVERRIDE CIRCUIT (PROTEUS SNIP)

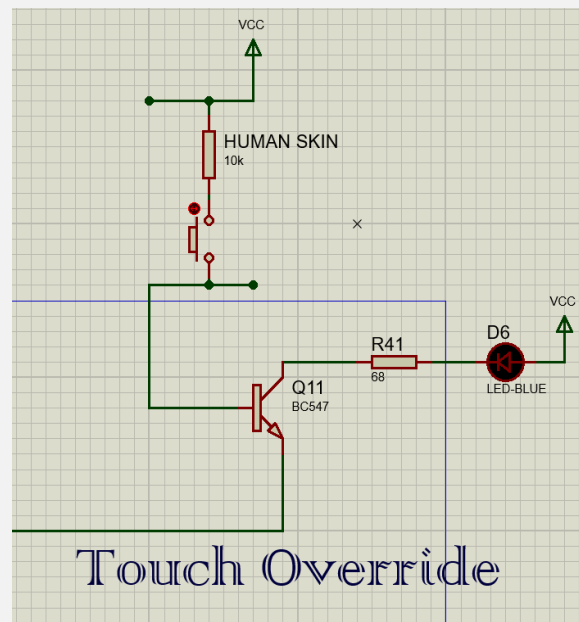
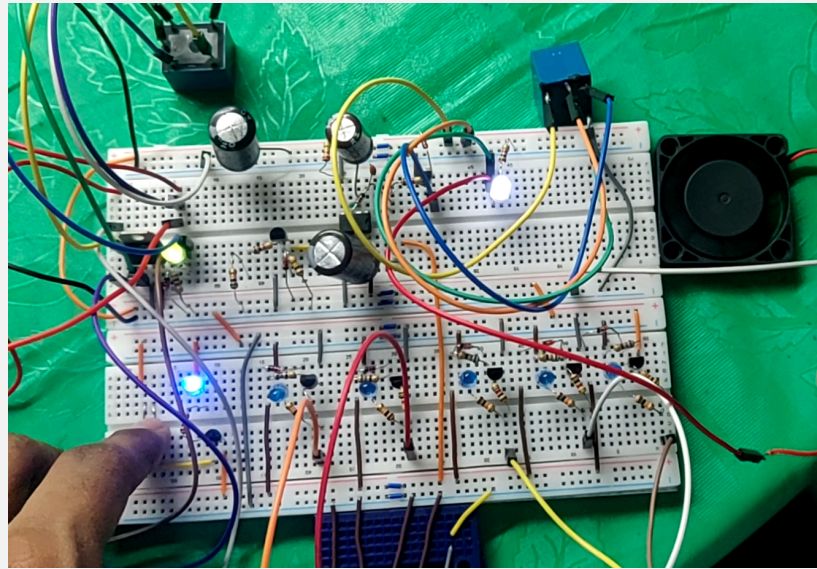


FIGURE: TOUCH-OVERRIDE CIRCUIT (HARDWARE PHOTO)



4.3 Uninterruptible Power Supply and Power Isolation Architecture

The system incorporates a dual-rail Smart UPS architecture that prioritizes logic preservation over heavy load consumption:

- Normal Operation (SW1 Closed / MAIN present): SW1 connects the 7.4V MAIN rail, through a series current-limiting resistor ("FAKE RES", standing in for real load/wiring resistance), to the coil of relay RL1, energizing it and throwing its SPDT contact to the MAIN side. This connects the regulator's input (node 1) to the 7.4V MAIN rail, while the same MAIN supply directly runs the 12V mini cooling fans and LED lighting modules. A flyback diode (D24) across the RL1 coil clamps the inductive kick generated when the relay later de-energizes.
- UPS Power Outage Mode (SW1 Open / Simulated Blackout): Opening SW1 removes coil current from RL1, so its spring-return throws the SPDT contact back to the BACKUP side, connecting node 1 to the 7.4V backup battery instead of the MAIN rail. RL1 is therefore doing double duty — it is both the sensor that detects whether main power is present and the switch that selects which source feeds the regulator, a classic changeover-UPS topology rather than a separate control signal driving the relay. Because the heavy 12V loads are fed from the MAIN rail directly rather than through the regulator, they drop out immediately on blackout, while the logic supply is preserved.
- Regulation and Output Smoothing: Whichever rail RL1 has selected (MAIN or BACKUP) feeds the LM7805 regulator's (U3) V_IN pin (node 1). The regulator's GND ties to the common ground (node 2), and its V_OUT (node 3) produces the regulated 5V logic rail (VCC) that powers the system's ICs and sensors. A 1000 μ F capacitor (C4) across the output smooths the rail and holds up the logic supply briefly during the switchover transient; a voltmeter probe across the output confirms VCC stays regulated regardless of which source is active. Note that the Proteus

schematic labels the two sources as 12V MAIN and 6V BACKUP for simulation convenience — in the actual build both rails are 7.4V, as stated elsewhere in this report, so the regulator's dropout margin should be evaluated against 7.4V on both inputs rather than the schematic's nominal 12V/6V labels.

- Backup Battery Isolation (SW2): A second KCD11 rocker switch is wired in series with the 7.4V backup battery rail. Opening SW2 disconnects the battery from the circuit entirely — so even if RL1 throws its contact to the BACKUP side during a blackout, no power reaches the regulator unless SW2 is closed. This allows the battery to be manually isolated, e.g. for safety during storage or maintenance.

FIGURE: UPS MODULE WITH DUAL ROCKER SWITCHES (PROTEUS)

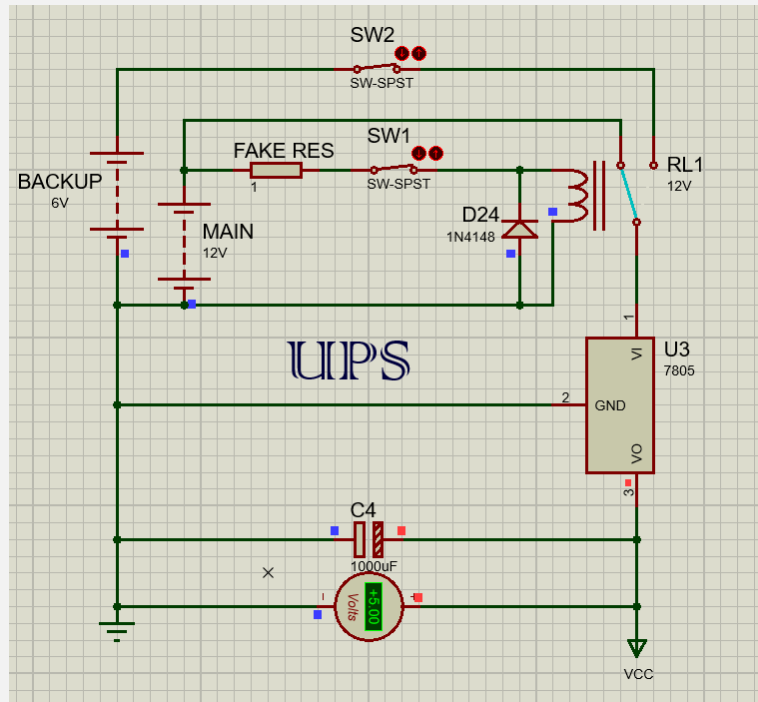
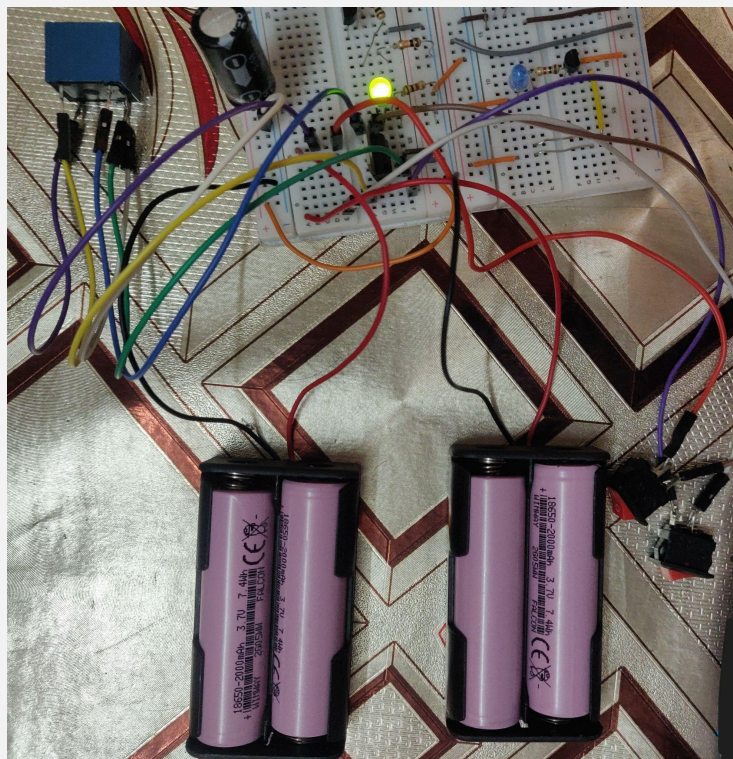


FIGURE: UPS MODULE WITH DUAL ROCKER SWITCHES (HARDWARE)



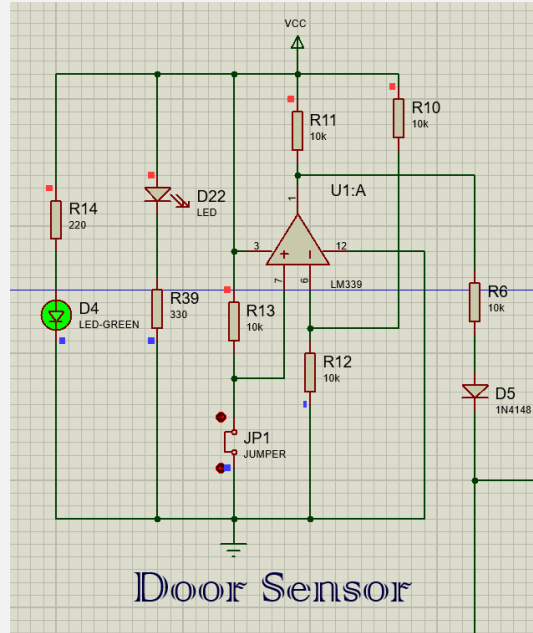
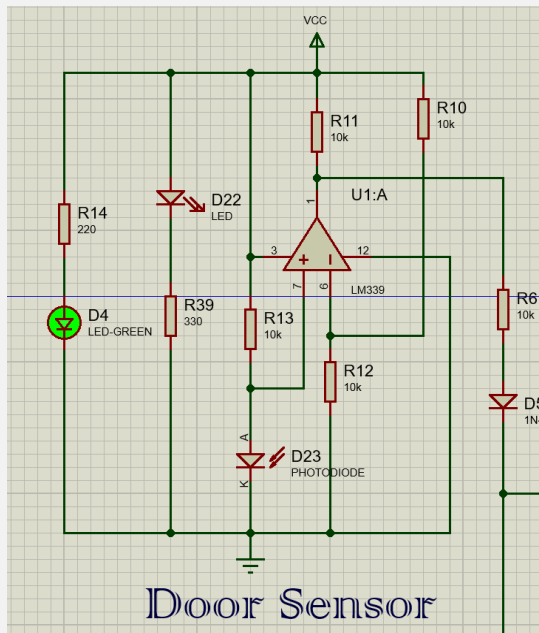
4.4 Infrared Door Sensor Module (Simulation Only, Not Implemented in Hardware — Comparator U1:A)

This module was laid out in Proteus around the first half of the LM339 quad comparator (U1:A) as an IR break-beam door sensor, but was left incomplete and was not carried into the physical breadboard build due to time constraints.

- IR Emitter Branch: D22 and current-limiting resistor R39 (330Ω) form the “transmitter” side, with VCC feeding D22 through R39 down to the sensing node. In finished hardware this represents a constantly-on IR LED shining across the doorway.
- Sensing Divider and Photodiode Stand-In: R13/R12 (10k/10k) form a voltage divider from the sensing node to ground, whose midpoint feeds the comparator. In real hardware a photodiode occupies this divider path, conducting when the beam is intact. Because Proteus always simulates total darkness, a real photodiode model would permanently report a broken beam with no way to toggle states during a demo — jumper JP1 stands in for it instead: closing JP1 mimics “beam intact,” opening it mimics “beam broken,” giving manual control for demonstration purposes.
- Reference Divider: R6 (10k) and D5 (1N4148), also returning to ground, form a second, fixed divider that sets the comparator's threshold voltage — the diode's forward drop gives a stable reference rather than a purely resistive one.
- Pull-Up: R11 (and R10, wired in parallel on the same VCC rail) pulls U1:A's open-collector output up to VCC whenever the comparator is not actively sinking it, since LM339 outputs can only pull low, not drive high.
- Power Indicator: D4 (green LED) and R14 form a standalone power-on indicator (VCC → R14 → D4 → ground) and are not part of the sensing logic.

Net behavior: toggling JP1 shifts the sensing-node voltage relative to the R6/D5 reference. Once it crosses that threshold, U1:A's output flips between pulled-low and released-high, forming the door open/closed (beam-broken/beam-intact) signal that would drive entry/exit logic in the complete design.

FIGURE: INFRARED DOOR SENSOR MODULE (PROTEUS)



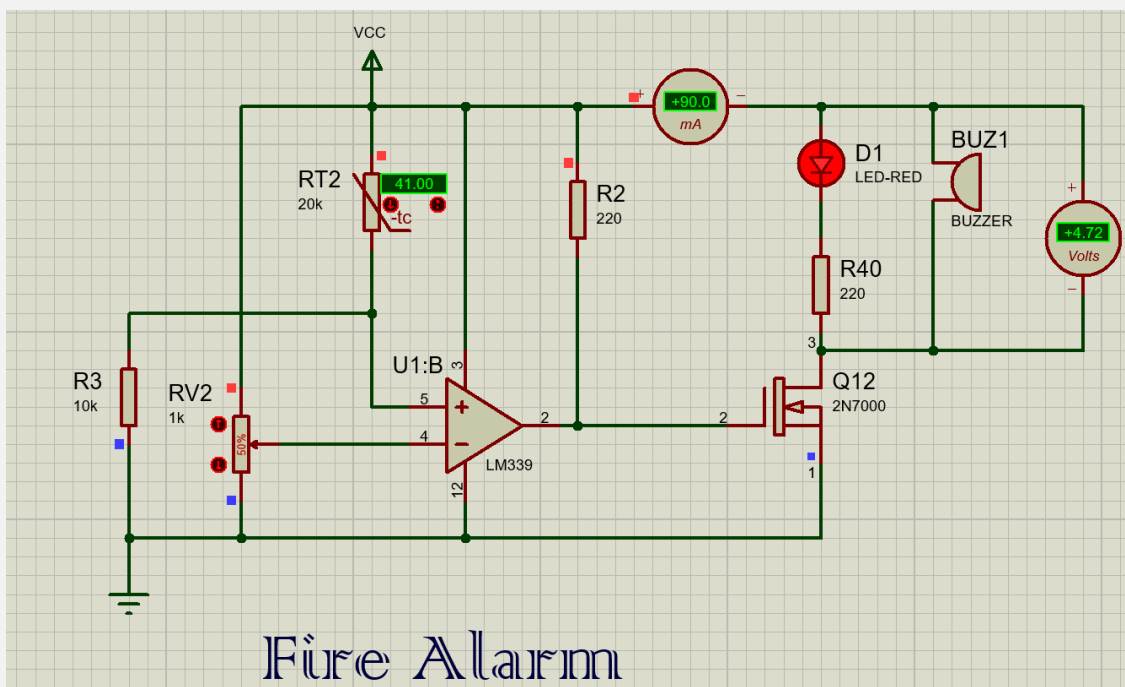
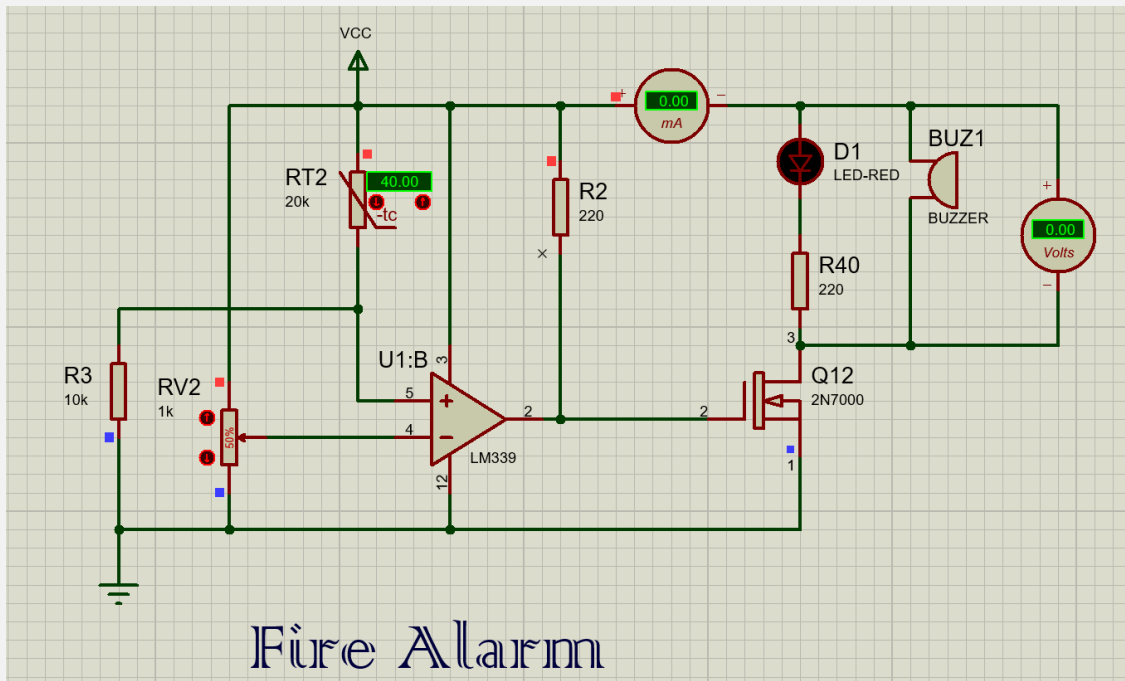
4.5 Thermistor-Based Fire Alarm Module (Simulation Only, Not Implemented in Hardware — Comparator U1:B)

This module was also laid out in Proteus, using the other half of the LM339 comparator (U1:B) as a temperature-triggered alarm, but like the door sensor it was left incomplete and was not carried into the physical breadboard build due to time constraints.

- **Temperature Sensing:** NTC thermistor RT2 (20k) and R3 (10k) form a voltage divider between VCC and ground feeding the comparator's non-inverting input (pin 5). RT2's resistance falls as temperature rises, so this node's voltage rises with temperature.
- **Threshold Setting:** Potentiometer RV2 (1k) forms a second divider between VCC and ground, with its wiper feeding the inverting input (pin 4). RV2 is calibrated so its wiper voltage corresponds to the desired trip temperature (approximately 40°C in the simulated setting).
- **Comparator Switching:** Below the threshold temperature, pin 5's voltage stays below pin 4's, so U1:B's open-collector output (pin 2) actively sinks current and stays low. R2 (220Ω) is the pull-up resistor on this output. Once the sensed temperature exceeds the RV2-set threshold, pin 5's voltage rises above pin 4's and the output releases, pulled high through R2.
- **Alarm Drive Stage:** The comparator output feeds the gate of Q12 (2N7000 MOSFET) through gate resistor R40 (220Ω). When the output goes high, Q12 turns on and pulls its drain low, completing the ground path for D1 (LED-RED) and BUZ1 (buzzer), both of which return to VCC through the top supply rail — so the LED lights and the buzzer sounds together.

Net behavior: once the sensed temperature rises above the RV2-calibrated threshold ($\sim 40^{\circ}\text{C}$), the comparator trips, Q12 switches on, and D1/BUZ1 activate together as a combined visual-and-audible fire alarm; both drop out again once the temperature falls back below the threshold. The ammeter and voltmeter probes shown in the schematic are simulation instruments only and are not part of the alarm's functional logic.

FIGURE: THERMISTOR-BASED FIRE ALARM MODULE (PROTEUS)



5. Hardware Troubleshooting

5.1 Component Failure Analysis: Defective BJT Batch

During physical breadboard assembly, the team encountered severe logic instability in the touch-override circuit section. Although the multi-transistor flip-flop design functioned flawlessly in Proteus simulation, the physical circuit exhibited random triggering, latch lockup, and failure to reset.

Extensive diagnostic isolation revealed that the root cause was a defective batch of physical BC547 BJT transistors possessing abnormal internal leakage current and inconsistent current gain (h_{FE}). Hours spent attempting to stabilize a mathematically correct schematic highlighted a critical practical lesson: real-world discrete semiconductors frequently depart from idealized simulation models.

5.2 Diagnostic Methodology and Circuit Optimization

To resolve the issue without jeopardizing project deadlines, the team executed a structured diagnostic process:

- **Component Isolation:** Transistors were removed from the breadboard and individually tested on a multimeter transistor socket, exposing collector-emitter leakage in 4 out of 6 units.
- **Circuit Redesign and Optimization:** The multi-transistor latch topology was replaced with an elegant single-BJT touch-bias arrangement. This reduced active component count, eliminated phantom switching, and achieved 100% trigger reliability.
- **Time and Resource Management:** By streamlining the touch override circuit, the team recovered lost troubleshooting time, enabling full assembly and testing of the remaining system modules (Occupancy and UPS).

6. Discussion

The Smart Classroom Project demonstrates a functional breadboard-based integration of automated occupancy-based load management, manual touch override, and uninterrupted power topology, assembled and wired across two interconnected breadboards. By combining mechanical microswitches, BC547-based occupancy sensing with NE555 delay-off timing, and 1-BJT touch logic, the prototype reduces wasted electrical energy in unoccupied seating positions while preserving critical logic operation during power interruptions. The IR-based door entry/exit sensor and the NTC thermistor fire alarm module from the original Proteus schematics design (from Sections 4.4 and 4.5) were, due to time constraints, not finished or carried over into the physical breadboard build.